

DESIGN AND ANALYSIS OF ALGORITHMS

Practical No. 1: Prime Number Generation and Complexity Analysis

1. Aim

To generate all prime numbers up to a given number n and analyze the time and space complexity of different approaches, including the Sieve of Eratosthenes.

2. Objective

- Understand time and space complexity.
- Compare simple prime checking with an optimized method.
- Implement the Sieve of Eratosthenes efficiently.

3. Theory

Prime number generation is a common example for studying algorithm efficiency. A simple method checks each number individually for divisibility. The Sieve of Eratosthenes improves efficiency by marking multiples of each prime as composite. Processing is required only up to the square root of n . The sieve has time complexity $O(n \log \log n)$ and space complexity $O(n)$.

4. Conclusion

The practical demonstrates how choosing a suitable algorithm can reduce computation. The Sieve of Eratosthenes is more efficient than checking every number independently, especially for large values of n .